5. All possible combination of NFA States have to be considered

2.

Given Statement

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Let P(n) be statement

$$P(n): 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Base Case:

$$n = 1$$

L.H.S

$$\sum_{i=1}^{1} i^2 = 1^2 = 1$$

R.H.S

$$\frac{n(n+1)(2n+1)}{6}$$

Sub $n = 1$

$$= \frac{1(1+1)(2(1)+1)}{6}$$

$= \frac{1 \times 2 \times 3}{6}$

$= 1$

$\therefore$ L.H.S = R.H.S

P(1) is true

$\therefore$ The Statement is true for $n = 1$

Now assume $n = k$

$$1^2 + 2^2 + 3^2 + \ldots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

To prove it for P(k+1) is true

L.H.S

$$1^2 + 2^2 + \ldots + k^2 + (k+1)^2$$

using induction

$$= \frac{k(k+1)(2k+1)}{6} + (k+1)^2$$

$$(k+1)\left[\frac{k(2k+1)}{6} + (k+1)\right]$$

$$= \frac{k+1\,(2k^2 + k + 6k + 6)}{6}$$

$$= \frac{(K+1)(2K^2+7K+6)}{6}$$

$$= \frac{(K+1)(K+2)(2K+3)}{6}$$

R.H.S

$n = K+1$

$$= \frac{n(n+1)(2n+1)}{6}$$

$$= \frac{(K+1)[(K+1)+1][2(K+1)+1]}{6}$$

$$= \frac{(K+1)(K+2)(2K+3)}{6}$$

Since

L.H.S = R.H.S

true

∴ By the principle of mathematical induction

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

Hence proved.